3.5 Inequations

Earlier we have learnt to construct linear equations. Let us now study about 'Inequations'.

As per the norms the minimum age limit to own a driving licence is 18 years. Therefore, when Rajiv owns a driving licence, we can say that he is *at least* 18 years of age.

Now, if Rajiv's age is represented by x, then this statement can be written mathematically as $x \geq 18$, that is, we are not sure about his age, still we can say that his age is greater than or equal to 18 years.

Similarly, the statement *'This jug can hold up to 5 litres of water'* can be written mathematically as $x \leq 5$, where 'x' represents the volume of water in the jug.

We know that, the sum of the measures of any two sides of a triangle is greater than the measure of its third side. Thus, if the measure of the three sides are represented by a, b and c units, then we may write this fact as $a + b > c$, $a + c > b$ and $b + c > a$.

When $x \neq 10$, then either $x > 10$ or $x < 10$. That is, if the value of the variable 'x' is not 10, then the value of 'x' will either be greater than 10 or be less than 10.

An algebraic statement that shows two algebraic expressions being unequal is known as an algebraic inequation.

In general, when two expressions are compared, one might be; less than (<), less than or equal to (≤), greater than (>), greater than or equal to (≥) the other.

In an inequation, the algebraic expressions are connected by one out of the four signs of inequalities, namely, >, ≥, < and ≤.

Try these

Construct inequations for the following statements:

1. Ramesh's salary is more than ₹25,000 per month.
2. This lift can carry maximum of 5 persons.
3. The exhibition will be there in town for at least 100 days.

3.5.1 Solving linear Inequations

A simple linear equation has atmost one solution, but a linear inequation may have many solutions.

To solve an inequation, it is necessary to know the set of values that the variable symbol can be substituted with. The collection of of all such values of an inequation is known as **solution** of the inequation.

For example, the solution of the equation $3x - 3 = 12$ is 5. (How?) Let us find the solution for the inequation $3x - 3 < 12$, where x is a natural number. Note that, the

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